INCH-POUND

MIL-S-20708/50F
13 March 1989

SUPERSEDING
MIL-S-20708/50E
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO TORQUE RECEIVER TRANSMITTER, TYPE 23TRX4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	720 0	milliamps	maximum
Primary Power	4 6	watts	maximum
Impedance			
Zss	6 5-8 1	ohms	min -max
Impedance Angle			
Zss	44 0-55 0	degrees	min -max
Transformation Ratio	0 783	-----	±2%
Phase Shift (Lead)	2 0	degrees	maximum
Electrical Error	6 0	minutes	maximum
Receiver Error	45 0	minutes	maximum
Null Voltage			
Total	150 0	millivolts	maximum
Fundamental	30 0	millivolts	maximum
Synchronizing Time			
30° ±2°	1 0	seconds	maximum
Synchronizing Time			
177° ±2°	2 0	seconds	maximum
Torque Gradient	0 25	ounce-inches per degree	minimum
Radial Play	0 0010	inches	maximum
End Play	0 0025	inches	±0 0020
Temperature Rise	30 0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	6 5	watts	maximum

(F) denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U + 0000 - 0002 Ⓕ
M20708/50-01A	676	719	3 160	Splined	Terminal Board	2405

1/Military part number M20708/50-01A shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS			NEW MILITARY PART NUMBERS Ⓕ
MIL-S-20708/50D	MIL-S-20708/50C	MIL-S-20708/50B	
M20708/50-001	M20708/50C-001	M20708/50-001 Ⓕ	M20708/50-01A

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians:
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI, AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC CC

Preparing Activity:
Navy-AS

(Project 5990-0349-33)

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